Chapter 10: Rotational Motion and Angular Momentum

T. K. Deskins, Ph.D.

Professor of Physics & Chemistry

Southern Union State Community College

PHY201

Lecture 10

Part I

Rotational Kinematics

Outline of Part I: Rotational Kinematics

Angular Quantities

Kinematics of Rotational Motion

Three New Symbols

Greek Symbols for Rotational Quantities

Symbol	**Name**	**Quantity**	**SI Unit**
θ	theta	angular position / displacement	radian (rad)
ω	omega	angular velocity	rad/s
α	alpha	angular acceleration	rad/s^2

These three symbols are the rotational analogues of x, v, and a.

Every linear kinematic and dynamic equation has a **direct rotational equivalent** — only the symbols change.

- $2\pi\,\text{rad} = 360° = 1\,\text{revolution}$
- Radians are *dimensionless* (ratio of arc length to radius), but we write "rad" to signal the context.

Angular Displacement $\Delta\theta$

Angular Displacement

$$\Delta\theta = \frac{\Delta s}{r} = \frac{\Delta x}{r}$$

where Δs is the arc length traveled and r is the radius. (SI: rad)

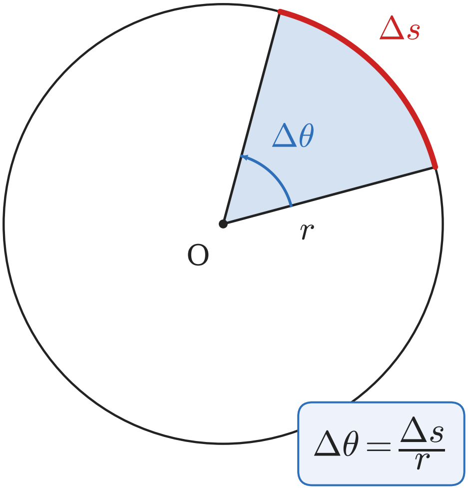
Rearranging:

$$\boxed{\Delta x = r\,\Delta\theta}$$

A point at distance r from the axis moves a linear distance $\Delta x = r\Delta\theta$ when the object rotates by $\Delta\theta$.

$\Delta\theta = 1\,\text{rad} \implies$ arc length equals the radius.

A full rotation: $\Delta\theta = 2\pi$ rad.



Angular Velocity ω

Angular Velocity

$$\omega = \frac{\Delta\theta}{\Delta t} \qquad \text{(SI: rad/s)}$$

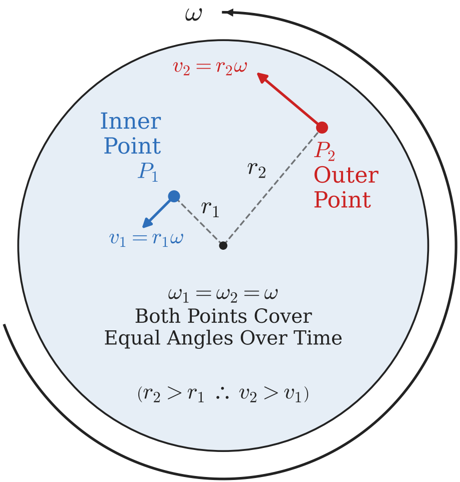
Relationship to Linear Speed

Since $\Delta x = r\Delta\theta$, dividing by Δt:

$$\frac{\Delta x}{\Delta t} = r\frac{\Delta\theta}{\Delta t}$$

$$\therefore \quad \boxed{v = r\omega}$$

Every point on the object has the *same* ω, but points farther from the axis move *faster*.



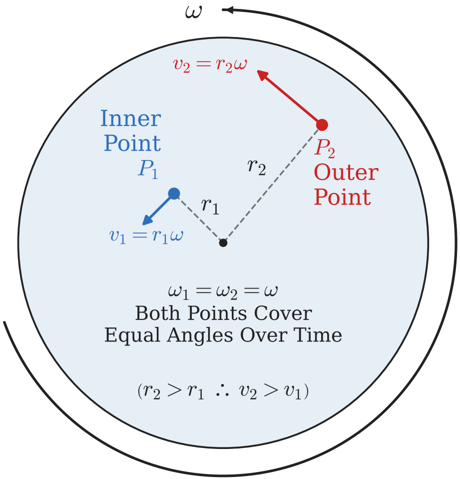
Conventional Direction of Angular Velocity ω

$$\omega > 0: \text{ counterclockwise (CCW)}$$
$$\omega < 0: \text{ clockwise (CW)}$$

📝 This should look familiar! It is the same convention we use for torque $\vec{\tau}$.
This direction is determined by the "*right-hand rule*".

💡 Using your **right hand**, curl your fingers in the rotation direction. Your thumb will point in the direction of $\vec{\omega}$.

🔬 **Why is CCW positive**? When your right-hand fingers curl CCW, your right-hand thumb points *toward you*.
This direction is the conventional $+z$ axis, just as $\rightarrow$ is the conventional $+x$ axis.



Angular Acceleration α

Angular Acceleration

$$\alpha = \frac{\Delta\omega}{\Delta t} \qquad (\text{SI: rad/s}^2)$$

Relationship to Tangential Acceleration

Recall $v = r\omega$. Seeing how they change over time gives:

$$\frac{\Delta v}{\Delta t} = r\frac{\Delta\omega}{\Delta t}$$

$$\therefore \quad \boxed{a_t = r\alpha}$$

a_t is the **tangential** (linear) acceleration of a point at radius r.

Do not confuse $a_t = r\alpha$ with centripetal acceleration:

$$a_c = \frac{v^2}{r} = r\omega^2$$

a_c points *inward*; a_t is *tangential*. Both can exist simultaneously.

Check Your Understanding

A merry-go-round completes one full revolution in 4.00 s.

(a) What is its angular velocity ω?

(b) A child sits 1.50 m from the center. What is the child's linear speed?

Come to lecture to see the answer.

Check Your Understanding

A spinning wheel slows from $\omega_0 = 10.0\,\frac{\text{rad}}{\text{s}}$ to $\omega_f = 2.00\,\frac{\text{rad}}{\text{s}}$ in $4.00\,\text{s}$.

(a) What is the angular acceleration?

(b) Is the wheel speeding up or slowing down? How do you know from the sign?

Come to lecture to see the answer.

Outline of Part I: Rotational Kinematics

Angular Quantities

Kinematics of Rotational Motion

Kinematic Equations: Linear $\longleftrightarrow$ Rotational

Linear	**Rotational**
$v(t) = v_0 + at$	$\omega(t) = \omega_0 + \alpha t$
$x(t) = x_0 + v_0 t + \frac{1}{2}at^2$	$\theta(t) = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$
$v^2 = v_0^2 + 2a(x - x_0)$	$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$
$v_{\text{avg}} = \dfrac{v_f + v_0}{2}$	$\omega_{\text{avg}} = \dfrac{\omega_f + \omega_0}{2}$

☠️ **Caution:** These equations are only valid when α and a are **constant**.

(🔬 If either acceleration changes over time, then calculus is required.)

Example: Merry-Go-Round — Kinematics

A merry-go-round (MGR) is pushed from rest by Bill, who applies a tangential force $F = 200\,\mathrm{N}$ at the rim ($R = 2.00\,\mathrm{m}$). The MGR has mass $M = 100\,\mathrm{kg}$. Using $\alpha = 2.00\,\frac{\mathrm{rad}}{\mathrm{s}^2}$, find ω_F, v_F, and $\Delta\theta$ after $t = 3.00\,\mathrm{s}$ starting from rest.

$$\omega_F = \omega_0 + \alpha t = 0 + (2.00)(3.00)$$

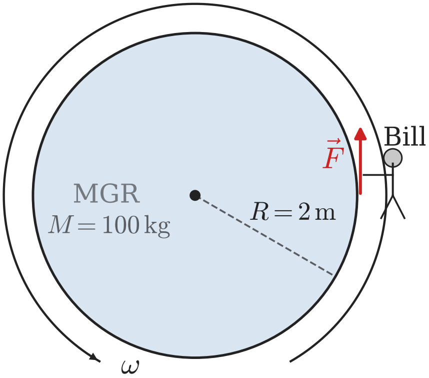
$$\boxed{\omega_F = 6.00\,\frac{\mathrm{rad}}{\mathrm{s}}}$$

$$v_F = R\omega_F = (2.00)(6.00)$$

$$\boxed{v_F = 12.0\,\frac{\mathrm{m}}{\mathrm{s}}}$$

$$\Delta\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2 = 0 + \tfrac{1}{2}(2.00)(9.00)$$

$$\boxed{\Delta\theta = 9.00\,\mathrm{rad} \approx 1.43\,\mathrm{rev}}$$



Check Your Understanding

A drill bit starts from rest and reaches $\omega_f = 628\,\frac{\text{rad}}{\text{s}}$ ($\approx 6000\,\text{rpm}$) in $0.500\,\text{s}$.

(a) What is the angular acceleration?

(b) How many revolutions does the bit complete while accelerating?

Come to lecture to see the answer.

Part II

Rotational Dynamics

Outline of Part II: Rotational Dynamics

Net Torque Causes Angular Acceleration

Linear (Newton's 2nd Law)

$$\vec{a} = \frac{1}{m}\vec{F}_{\text{net}}$$

$$\sum \vec{F} = m\vec{a}$$

Net **force** causes linear acceleration.
m measures **resistance to linear acceleration**.

Rotational Analog

$$\vec{\alpha} = \frac{1}{I}\vec{\tau}_{\text{net}}$$

$$\sum \vec{\tau} = I\vec{\alpha}$$

Net **torque** causes angular acceleration.
I measures **resistance to angular acceleration**.

The **moment of inertia** I plays the same role in rotation that **mass** m plays in linear motion: it measures **how hard it is to change an object's *rotational* state of motion**.

Moment of Inertia: Definition

Moment of Inertia (I) — SI Unit: kg·m^2

For a collection of point masses:

$$I = \sum m_i r_i^2 \qquad (\text{SI: kg}\cdot\text{m}^2)$$

where r_i is the *perpendicular distance* from the rotation axis to mass m_i.

- I is like mass, but **also depends on where the mass is located** relative to the axis.
- Mass far from the axis contributes *more* to I (by r^2).
- $\therefore$ you can reduce I by moving mass closer to the axis (e.g., a figure skater pulling in their arms).

For a **point mass** at radius R from the axis: $I_{\text{PM}} = MR^2$

Moments of Inertia for Common Shapes

Geometry	I
Hoop about cylinder axis	MR^2
Solid cylinder about cylinder axis	$MR^2/2$
Thin rod about center, $\perp$ to L	$ML^2/12$
Solid sphere about any diameter	$2MR^2/5$
Hoop about any diameter	$MR^2/2$
Annulus about cylinder axis	$M(R_1^2 + R_2^2)/2$
Solid cylinder about midpoint $(L/2)$	$MR^2/4 + ML^2/12$
Thin rod about end, $\perp$ to L	$ML^2/3$
Thin spher. shell about any diam.	$2MR^2/3$
Slab about perpendicular axis	$M(a^2 + b^2)/12$

For the same total mass M and radius R, a **hoop** (MR^2) is harder to spin than a **solid disk** $(MR^2/2)$.

This is because the hoop's mass is concentrated **at the rim**, farther from the axis, while the solid disk's mass is concentrated **halfway between the rim and the center**.

The lever arm for the hoop $r_\perp$ is twice as long as the solid disk's $r_\perp$,

$$\therefore \quad I_{\text{hoop}} = 2I_{\text{solid disk}}.$$

Parallel-Axis Theorem

Parallel-Axis Theorem

If I_{CM} is the moment of inertia about an axis through the center of mass, then the moment of inertia about any *parallel* axis displaced by distance d is:

$$I = I_{\mathrm{CM}} + Md^2$$

- Useful when combining objects: treat each piece separately, then add.
- A **point mass** has $I_{\mathrm{CM}} = 0$, so $I_{\mathrm{PM}} = 0 + MR^2 = MR^2$.
  (This matches the point-mass formula on the equation sheet.)
- Example: a person (treated as point mass M_c) riding a merry-go-round at radius R_c:
  $I_{\mathrm{person}} = M_c R_c^2$

Example: Merry-Go-Round — Finding α

A solid-cylinder merry-go-round ($M = 100\,\mathrm{kg}$, $R = 2.00\,\mathrm{m}$) starts from rest. Bill pushes with $F = 200\,\mathrm{N}$ tangentially at the rim. Find α.

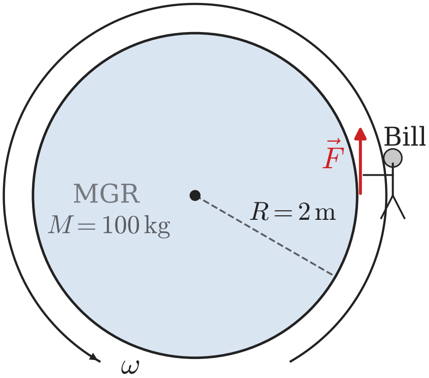
$$I = I_{\mathrm{cyl}} = \frac{1}{2}MR^2$$

$$\tau_{\mathrm{net}} = RF = I\alpha$$

$$\alpha = \frac{\tau_{\mathrm{net}}}{I} = \frac{RF}{\frac{1}{2}MR^2} = \frac{2F}{MR}$$

$$= \frac{2(200)}{(100)(2.00)}$$

$$\boxed{\alpha = 2.00\,\frac{\mathrm{rad}}{\mathrm{s}^2}}$$



Example: MGR + Cynthia — System Moment of Inertia

Now Cynthia ($M_c = 30\,\mathrm{kg}$) stands $R_c = 1\,\mathrm{m}$ from the center of the MGR (at rest). Bill again applies $F = 200\,\mathrm{N}$ at the rim ($R = 2.00\,\mathrm{m}$). Find the new α.

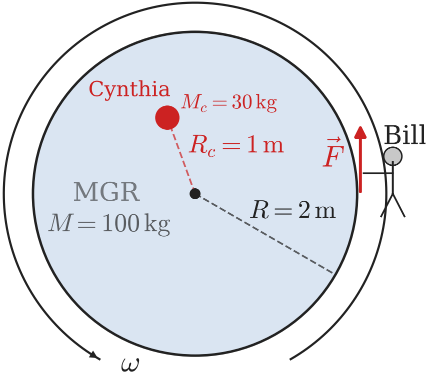
$$I_{\text{sys}} = I_{\text{MGR}} + I_{\text{Cynthia}} \quad \text{(Parallel-Axis Theorem)}$$

$$= \frac{1}{2}MR^2 + M_c R_c^2$$

$$\tau_{\text{net}} = RF = I\alpha$$

$$\alpha = \frac{\tau_{\text{net}}}{I} = \frac{RF}{\frac{1}{2}MR^2 + M_c R_c^2}$$

$$\boxed{\alpha = 1.74\,\frac{\mathrm{rad}}{\mathrm{s}^2}}$$



Check Your Understanding

Two objects have the same mass M and outer radius R: a solid disk and a thin-walled hollow cylinder (hoop).

The same torque is applied to each.

Which has the greater angular acceleration, and why?

Come to lecture to see the answer.

Check Your Understanding

A figure skater spins with arms extended ($\omega_0 = 1.00\,\frac{\text{rad}}{\text{s}}$, $I_0 = 4.50\,\text{kg}\cdot\text{m}^2$). She then pulls her arms in so that $I_f = 1.50\,\text{kg}\cdot\text{m}^2$.

What is her new angular velocity?

(Hint: no net torque acts, so $\sum\tau = 0$ and $\Delta L = 0$.)

Come to lecture to see the answer.

Outline of Part II: Rotational Dynamics

Rotational Kinetic Energy

Kinetic Energy of Rotation

A rotating object has kinetic energy even if its center of mass is not moving:

$$KE_{\text{rot}} = \frac{1}{2} I\omega^2$$

Compare: $KE_{\text{linear}} = \frac{1}{2}mv^2 \quad (I \leftrightarrow m,\ \omega \leftrightarrow v)$

Rolling Objects: Total KE

An object rolling without slipping has both translational and rotational KE:

$$KE_{\text{total}} = \frac{1}{2}mv_{\text{CM}}^2 + \frac{1}{2}I\omega^2$$

Rolling condition (no slipping): $v_{\text{CM}} = R\omega$

Work-Energy Theorem for Rotation

Rotational Work-Energy Theorem

$$W_{\text{net}} = \tau_{\text{net}}\,\Delta\theta = \Delta KE_{\text{rot}} = \frac{1}{2}I(\omega_f^2 - \omega_i^2)$$

Compare: $W_{\text{net}} = F_{\text{net}}\,\Delta x = \Delta KE = \frac{1}{2}m(v_f^2 - v_i^2)$

Finding $\Delta\theta$ from ω_f (start from rest)

Setting $\omega_i = 0$:

$$\tau\,\Delta\theta = \frac{1}{2}I\omega_f^2$$

$$\Delta\theta = \frac{I\omega_f^2}{2\,\tau}$$

For the MGR:

$$\Delta\theta = \frac{I\omega_f^2}{2\tau} = \frac{(200)(6.00)^2}{2(400)}$$

$$= \frac{7200}{800} = 9.00\,\text{rad} \checkmark$$

Example: Work-Energy for MGR + Cynthia

Using the Work-Energy Theorem, find $\Delta\theta$ for the MGR + Cynthia system.
Recall: $I_{\text{sys}} = 230\,\text{kg}\cdot\text{m}^2$, $\tau = 400\,\text{N}\cdot\text{m}$, $\omega_f = 5.22\,\frac{\text{rad}}{\text{s}}$.

$$\Delta\theta = \frac{I_{\text{sys}}\,\omega_f^2}{2\,\tau}$$
$$= \frac{\left(\frac{1}{2}MR^2 + M_cR_c^2\right)\omega_f^2}{2RF}$$
$$= \frac{(230)(5.22)^2}{2(400)}$$
$$\boxed{\Delta\theta = 7.83\,\text{rad}}$$

Compared with previous ex.($\Delta\theta = 9.00$ rad): Cynthia increased I, which slowed both α and $\omega_f \Rightarrow$ fewer radians traversed in the same time.

Check:
$W = \tau\Delta\theta = (400)(7.83) = 3130\,\text{J}$
$\frac{1}{2}I\omega_f^2 = \frac{1}{2}(230)(27.2) = 3130\,\text{J}$ ✓

Example: Rolling Sphere Down an Incline

A solid sphere ($M = 5.00\,\mathrm{kg}$, $R = 0.200\,\mathrm{m}$) rolls without slipping from rest down an incline of vertical height $h = 2.00\,\mathrm{m}$. Find its center-of-mass speed at the bottom.

Energy conservation ($KE_i = 0$, $PE_f = 0$):

$$\begin{aligned} mgh &= \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \\ &= \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mR^2\right)\left(\frac{v}{R}\right)^2 \\ &= \frac{1}{2}mv^2 + \frac{1}{5}mv^2 = \frac{7}{10}mv^2 \end{aligned}$$

$$v = \sqrt{\frac{10gh}{7}} = \sqrt{\frac{10(9.80)(2.00)}{7}}$$

$$\boxed{v = 5.29\,\frac{\mathrm{m}}{\mathrm{s}}}$$

Compare with a *sliding* (no friction) block:
$v_{\mathrm{slide}} = \sqrt{2gh} = 6.26\,\frac{\mathrm{m}}{\mathrm{s}}$
Rolling is **slower** because some energy goes into rotation rather than translation.

For a hoop ($I = mR^2$):
$v = \sqrt{gh} = 4.43\,\frac{\mathrm{m}}{\mathrm{s}}$ (slowest).

Check Your Understanding

A solid cylinder and a hoop have the same mass and radius. They are released from rest at the top of the same incline.

Which reaches the bottom first? Explain using the work-energy theorem without doing any calculation.

Come to lecture to see the answer.

Part III

Angular Momentum

Outline of Part III: Angular Momentum

Angular Momentum: Definition

Angular Momentum $\left(\vec{L}\right)$ — SI Unit: kg·m²/s

For a rigid body rotating about a fixed axis:

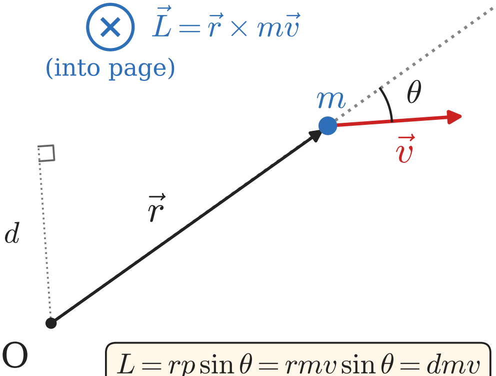
$$\vec{L} = I\vec{\omega} \qquad (\text{SI: kg·m}^2/\text{s})$$

For particle at position $\vec{r}$ with momentum $\vec{p}$:

$$\vec{L} = \vec{r} \times \vec{p} \qquad \Rightarrow \qquad L = rp\sin\theta_{\vec{r}\vec{p}}$$

Analogy with Linear Momentum

Linear	Rotational
$\vec{p} = m\vec{v}$	$\vec{L} = I\vec{\omega}$
m	I
$\vec{v}$	$\vec{\omega}$



$$\boxed{L = rp\sin\theta = rmv\sin\theta = dmv}$$

A moving particle has angular momentum about any point not on its line of motion — even if it's not rotating!

Newton's Second Law for Rotation

Linear

$$\vec{F}_{\text{net}} = \frac{\Delta \vec{p}}{\Delta t}$$

Net force $\Rightarrow$ rate of change of linear momentum.

If $\vec{F}_{\text{net,ext}} = 0$:

$$\implies \Delta \vec{p} = 0 \text{ (conservation of } \vec{p}\text{)}$$

Rotational Analog

$$\vec{\tau}_{\text{net}} = \frac{\Delta \vec{L}}{\Delta t}$$

Net torque $\Rightarrow$ rate of change of angular momentum.

If $\vec{\tau}_{\text{net,ext}} = 0$:

$$\implies \Delta \vec{L} = 0 \text{ (conservation of } \vec{L}\text{)}$$

Conservation of Angular Momentum: When the *net external torque* on a system is zero, the total angular momentum remains constant:

$$I_i \omega_i = I_f \omega_f \qquad (\text{if } \tau_{\text{ext}} = 0)$$

Conservation of Angular Momentum: Everyday Examples

Figure Skater

Arms out $\Rightarrow$ large I, small ω
Arms in $\Rightarrow$ small I, large ω

$L = I\omega = \text{const}$

The ice exerts almost no torque on the skater $\Rightarrow \tau_{\text{ext}} \approx 0$.

Diver

Tuck (small I) $\Rightarrow$ fast spin
Layout (large I) $\Rightarrow$ slow spin

Once airborne, $\tau_{\text{ext}} = 0$ (gravity acts at CM, producing no torque about CM).

In both cases: no external torque $\Rightarrow$ L is conserved, but the athlete can change I by redistributing their own mass $\Rightarrow$ ω changes to compensate.

Example: Angular Impulse on a Cylinder

A tangential force $F = 2.50\,\mathrm{N}$ is applied at the rim of a solid cylinder ($M = 4.00\,\mathrm{kg}$, $R = 0.260\,\mathrm{m}$) for $\Delta t = 0.150\,\mathrm{s}$. The cylinder starts from rest ($\omega_0 = 0$). Find ω_f.

Method 1: Angular Impulse-Momentum

$$\tau_{\text{net}}\,\Delta t = \Delta L = I\,\Delta\omega$$

$$RF\,\Delta t = \frac{1}{2}MR^2\,\omega_f$$

$$\omega_f = \frac{2F\,\Delta t}{MR} = \frac{2(2.50)(0.150)}{(4.00)(0.260)}$$

$$\boxed{\omega_f = 0.721\ \tfrac{\mathrm{rad}}{\mathrm{s}}}$$

Method 2: α then Kinematics

$$\alpha = \frac{\tau}{I} = \frac{RF}{\frac{1}{2}MR^2} = \frac{2F}{MR}$$

$$\omega_f = \omega_0 + \alpha\,\Delta t = 0 + \frac{2F}{MR}\Delta t$$

$$\boxed{\omega_f = 0.721\ \tfrac{\mathrm{rad}}{\mathrm{s}}}\ \checkmark$$

Both methods give the same answer — use whichever is simpler for the given information.

Check Your Understanding

A student sits on a frictionless rotating stool holding a spinning bicycle wheel with its axis vertical (ω_{wheel} upward). She then flips the wheel so the axis points downward.

What happens to the student, and why?

Come to lecture to see the answer.

Check Your Understanding

A solid disk ($I = 0.800\,\mathrm{kg}\cdot\mathrm{m}^2$) rotates at $\omega_0 = 5.00\,\frac{\mathrm{rad}}{\mathrm{s}}$. A net torque of $-1.20\,\mathrm{N}\cdot\mathrm{m}$ acts on it for $2.00\,\mathrm{s}$.

(a) What is the final angular velocity?

(b) What is the change in angular momentum?

Come to lecture to see the answer.

Outline of Part III: Angular Momentum

Angular Momentum and Its Conservation

Collisions of Extended Bodies

Gyroscopic Effects: Vector Aspects of Angular Momentum

Angular Momentum in Collisions

When an object *collides with* or *attaches to* a rotating body, angular momentum is conserved if no external torque acts:

$$L_i = L_f \qquad \Rightarrow \qquad I_i\,\omega_i + m\,v\,r = (I_i + mr^2)\,\omega_f$$

where r is the distance from the collision point to the rotation axis and mv is the linear momentum of the incoming object.

- The collision is typically **perfectly inelastic** (objects stick together).
- Kinetic energy is *not* conserved (some is lost to deformation/heat).
- Angular momentum *is* conserved (if $\tau_{\text{ext}} = 0$).

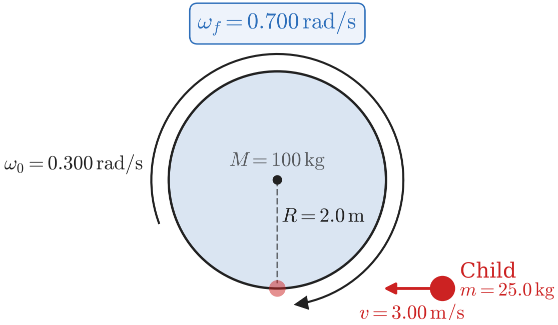
Example: Child Jumps onto Merry-Go-Round

A child ($m = 25.0\,\mathrm{kg}$) runs at $v = 3.00\,\frac{\mathrm{m}}{\mathrm{s}}$ and jumps onto the rim of a merry-go-round ($M = 100\,\mathrm{kg}$, $R = 2.00\,\mathrm{m}$) rotating at $\omega_0 = 0.300\,\frac{\mathrm{rad}}{\mathrm{s}}$ in the same direction. Find ω_f.

$$\begin{aligned} L_i &= L_{\mathrm{MGR}} + L_{\mathrm{child}} \\ &= I_{\mathrm{MGR}}\omega_0 + mvR \\ &= \frac{1}{2}(100)(2)^2(0.3) + (25)(3)(2) \\ &= 60 + 150 = 210\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s} \end{aligned}$$

$$I_f = I_{\mathrm{MGR}} + mR^2 = 300\,\mathrm{kg}\cdot\mathrm{m}^2$$

$$\boxed{\omega_f = \frac{L_i}{I_f} = \frac{210}{300} = 0.700\,\frac{\mathrm{rad}}{\mathrm{s}}}$$



The MGR sped up from 0.300 to 0.700 $\frac{\mathrm{rad}}{\mathrm{s}}$. Is KE conserved?

$$KE_i = \tfrac{1}{2}(200)(0.300)^2 + \tfrac{1}{2}(25)(3.00)^2 = 9.0 + 112.5 = 121.5\,\mathrm{J}$$

$$KE_f = \tfrac{1}{2}(300)(0.700)^2 = 73.5\,\mathrm{J}$$

$\boxed{\text{No — 48.0 J lost to deformation.}}$

Check Your Understanding

A solid disk ($I_0 = 0.500\,\mathrm{kg}\cdot\mathrm{m}^2$, $\omega_0 = 4.00\,\frac{\mathrm{rad}}{\mathrm{s}}$) is spinning freely. A small clay lump ($m = 0.100\,\mathrm{kg}$) is dropped and sticks to the rim at $R = 0.400\,\mathrm{m}$ from the center.

Find the new angular velocity ω_f.

Come to lecture to see the answer.

Outline of Part III: Angular Momentum

Angular Momentum as a Vector: The Right-Hand Rule

Direction of $\vec{L} = I\vec{\omega}$

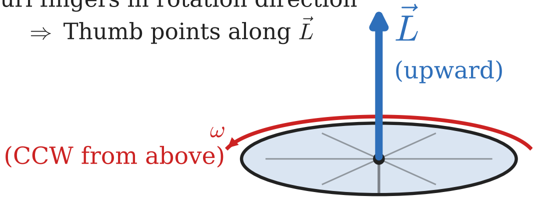
$\vec{L}$ is a vector. Its direction is given by the **right-hand rule**:

- Curl the fingers of your right hand in the direction of rotation.
- Your thumb points in the direction of $\vec{\omega}$ (and $\vec{L}$).

Right-Hand Rule:

Curl fingers in rotation direction

$\Rightarrow$ Thumb points along $\vec{L}$



Recall: $\vec{\tau} = \vec{r} \times \vec{F}$ is also a vector.

$$\vec{\tau}_{\text{net}} = \frac{\Delta \vec{L}}{\Delta t}$$

So $\vec{\tau}$ determines which way $\vec{L}$ changes — in both *magnitude* and *direction*.

Gyroscopes: Rotational Stability

Gyroscopic Stability

A rapidly spinning gyroscope **resists changes in the direction** of its spin axis, because a large $\vec{L}$ requires a large torque to change direction.

$$\vec{\tau}_{\text{net}} = \frac{\Delta \vec{L}}{\Delta t} \qquad \Rightarrow \qquad \Delta \vec{L} = \vec{\tau}_{\text{net}}\, \Delta t$$

For large $|\vec{L}|$ and small Δt: the angular change $\Delta \vec{L}/|\vec{L}|$ is tiny.

Applications of Gyroscopic Stability

- **Bicycle wheel**: spinning wheel resists tipping $\Rightarrow$ easier to balance.
- **Spinning bullet/football**: rifling or spiral pass stabilizes the trajectory.
- **Spacecraft attitude control**: reaction wheels change spacecraft orientation without fuel.
- **Navigation gyroscopes**: maintain fixed orientation for aircraft and ships.

Gyroscopic Precession

Precession

When a torque $\vec{\tau}$ acts *perpendicular* to $\vec{L}$ (e.g., gravity on a tilted top), the spin axis itself rotates — this is called **precession**.

The spin axis traces a **cone** around the vertical.

Precession angular velocity:

$$\Omega = \frac{\tau}{L} = \frac{Mgr}{I\omega}$$

where r is the distance from the support to the CM.

Counterintuitively, applying a downward torque does NOT push the axis downward — it causes it to *rotate sideways*.

The faster the spin (ω), the **slower** the precession (Ω) — a fast gyroscope is more stable.

Check Your Understanding

A bicycle wheel is held by a rope attached to one end of its axle. The wheel spins rapidly. When released, instead of falling, the wheel precesses horizontally.

(a) What is the direction of $\vec{L}$ for the spinning wheel (assume it spins CCW viewed from the right)?

(b) Gravity pulls the far end of the axle downward, creating a torque. What direction is this torque?

(c) Using $\Delta\vec{L} = \vec{\tau}\,\Delta t$, which way does the wheel precess?

Come to lecture to see the answer.

Check Your Understanding

Is it possible for an object to have angular momentum without spinning? Give an example or explain why not.

Come to lecture to see the answer.

Chapter Summary

Key Equations

- $v = r\omega; \quad a_t = r\alpha$
- $\omega(t) = \omega_0 + \alpha t$
- $\theta(t) = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$
- $\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$
- $\sum \tau = I\alpha$
- $KE_{\text{rot}} = \frac{1}{2}I\omega^2$
- $W_{\text{net}} = \tau_{\text{net}}\Delta\theta$
- $\vec{L} = I\vec{\omega}; \quad L = rp\sin\theta$
- $\tau_{\text{net}} = \Delta L/\Delta t$

Key Concepts

- All linear kinematic/dynamic equations have rotational analogues ($x \to \theta$, $v \to \omega$, $a \to \alpha$, $m \to I$, $F \to \tau$, $p \to L$).
- Kinematic equations require *constant* α.
- I depends on mass *and* its distribution.
- L is conserved when $\tau_{\text{ext}} = 0$ (figure skater, diver).
- A torque perpendicular to $\vec{L}$ causes precession.